Project Report

Credit Card Default Prediction

Using Classification and Risk-Based Techniques

Submitted to : Finance Club, IIT Roorkee

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1 Project Overview

1.1 What is Credit Card Default and Why is it Important?

Credit card default occurs when a customer is unable to repay their outstanding credit card balance. This poses significant financial risk to banks and lending institutions. Predicting which customers are likely to default in the upcoming month helps minimize losses, manage credit exposure, and take proactive action. Using historical repayment behavior, credit usage, and demographic data, our goal is to build a machine learning model that flags high-risk customers before they default. This enables smarter credit decisions and strengthens financial stability for lenders.

1.2 Business Objective

The primary objective of this project is to assist financial institutions in minimizing credit risk by identifying customers who are likely to default on their credit card payments in the upcoming month. Default risk is a critical factor in credit-based systems, and proactively managing it allows lenders to reduce financial losses, optimize credit allocation, and maintain portfolio health.

To address this, we apply classification-based machine learning techniques combined with risk-aware strategies such as threshold tuning, recall-focused evaluation F score, and feature engineering aligned with financial risk indicators (e.g., credit utilization, delay streaks, payment-to-bill ratios). Our model aims not only to predict defaulters accurately but to prioritize catching high-risk cases, even at the cost of occasional false alarms. This aligns with the core business goal: minimizing missed defaulters while maintaining operational efficiency.

1.3 Given Data

The dataset provided contains historical credit and repayment records for a large number of credit card customers. Each row represents a unique customer, and the columns describe both demographic attributes and financial behavior over a six-month period.

Key columns include:

- **LIMIT_BAL** – The credit limit assigned to the customer.
- **SEX, EDUCATION, MARRIAGE, AGE** – Basic demographic information.
- **BILL_AMT1 to BILL_AMT6** – Monthly bill statements from the past six months.
- **PAY_AMT1 to PAY_AMT6** – Monthly repayment amounts over the same six months.
- **PAY_0 to PAY_6** – Delay status of payments for each month (e.g., -1 = paid in advance, 0 = on time, 1 = one month delay, etc.).
- **next_month_default** – The target variable indicating whether the customer defaulted on payment in the following month (1 = default, 0 = no default).

The training dataset contains labeled records used to train and validate the model, while the final test (validation) set consists of similar customer records without the target variable. Our task is to accurately predict the default status for this validation set.

1.4 Problem Statement

The objective of this project is to develop a machine learning model that can accurately predict whether a credit card customer will default on their payment in the upcoming month. The prediction must be made based on historical customer behavior, including credit utilization, repayment history, and demographic factors.

This is a binary classification task where the goal is to identify high-risk customers before they default. By accurately forecasting defaults, financial institutions can proactively manage credit risk, reduce potential losses, and make more informed lending decisions. The model must be optimized not just for accuracy, but for recall and F_2-score, in order to prioritize catching defaulters even at the cost of occasional false positives.

1.5 Approach

The solution to the credit card default prediction problem was designed using a structured pipeline of data analysis, transformation, modeling, and evaluation. The approach followed these major steps:

- **Step 1: Data Exploration** – We began with exploratory data analysis to understand the distribution of features, identify patterns between customer attributes and default behavior, and check for class imbalance or missing values.
- **Step 2: Feature Engineering** – Based on domain understanding, we created new features such as credit utilization, payment-to-bill ratios, repayment delay streaks, and payment trends over time. These features aimed to capture customer behavior more effectively than the raw variables.
- **Step 3: Addressing Class Imbalance** – The dataset had far fewer defaulters compared to non-defaulters. We experimented with techniques like SMOTE and class weights to ensure the model paid sufficient attention to the minority class.
- **Step 4: Model Training and Evaluation** – We trained and compared multiple models including Logistic Regression, Random Forest, LightGBM, and XGBoost. Each model was evaluated using validation metrics such as F_2-score, recall, and AUC-ROC.
- **Step 5: Threshold Optimization** – Since our objective prioritized recall, we tuned the prediction threshold for each model to maximize the F_2-score, which weights recall higher than precision.
- **Step 6: Feature Selection and Finalization** – Using feature importance metrics from tree-based models, we removed low-impact features to reduce noise and improve generalization.
- **Step 7: Final Prediction** – The best-performing model (XGBoost with tuned threshold and selected features) was used to generate predictions for the unseen test dataset.

2 Data Preprocessing

Before training the models, several preprocessing steps were performed to clean and prepare the data:

- **Missing Values:** A total of 126 missing values were found in the `AGE` column. These were imputed using the median age of the dataset to preserve distribution symmetry and avoid introducing bias.
- **Inconsistent Categorical Entries:** Several categorical columns contained invalid or inconsistent values:
  - Values of `0` in the `marriage` column were replaced with `3 (Others)`, as 0 is not a valid marital status code.
  - Values `0, 5, 6` in the `education` column were also inconsistent. These were grouped and replaced with `4 (Others)`. Since these values accounted for over 1% of the data (311 entries), they were corrected rather than removed.
- **Incorrect Values in Precomputed Columns:** Two columns present in the original dataset contained invalid or inconsistent negative values:
  - `AVG_Bill_amt`: Although this column was provided in the dataset, some values were negative, which is not valid in the context of billing amounts. These incorrect entries were corrected by recalculating the average manually using `BILL_AMT1` to `BILL_AMT6`.
  - `PAY_TO_BILL_ratio`: This column, also present in the dataset, contained some negative values that are not meaningful in a financial context. These were clipped to zero to ensure the ratio remained within a valid range.

3 Exploratory Data Analysis (EDA)

3.1 Demographic Analysis

Demographic features such as marriage, sex, education, and age were analyzed to understand their distribution and relationship with default behavior.

- `marriage`: The dataset contains three categories: 1 (married), 2 (single), and 3 (others). distrubution shows that precentage of defaulters is slightly higher for married customers across all three categories.
- `sex`: Plot says that the percentage of defaulters is slightly higher for Female customers compared to male customers.
- `education`: Here education is categorized into 1 (graduate school), 2 (university), 3 (high school), and 4 (others). The distribution shows that the percentage of defaulters is highest for customers with High School education, followed by those with University and then Graduate School education. The "Others" category has the lowest default rate.
- `age`: Initially plotted box-plot of age indicated some outliers, which we handled by clipping with IQR method. Then KDE plot of age against the target variable `next_month_default` shows that the percentage of defaulters is highest for customers in the age group of 25-35 years, and it gradually decreases for older age groups. This suggests that younger customers are more likely to default compared to older customers.

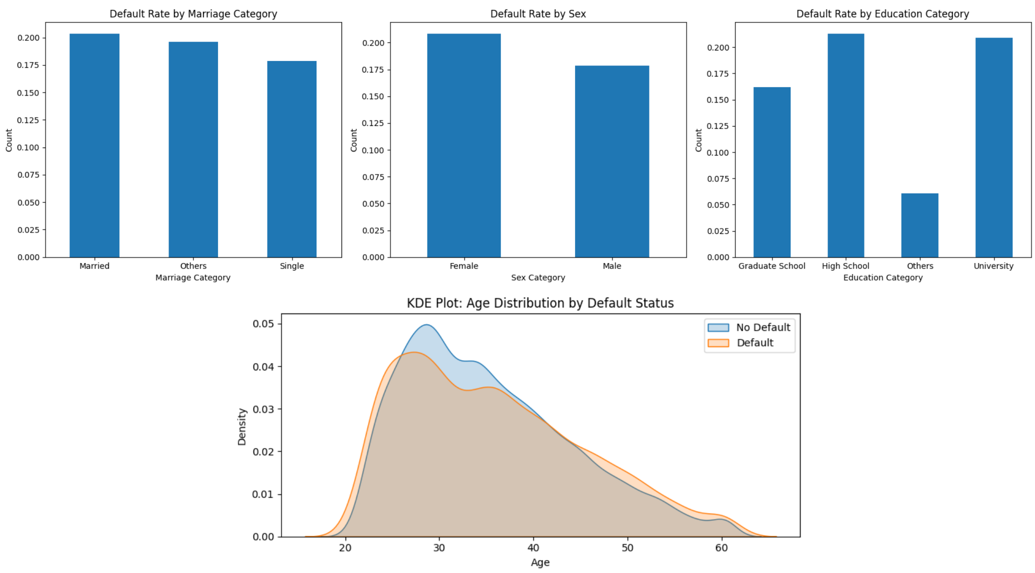

Figure 1: Distributions of Age, Marriage, Sex, and Education against Default Status

3.2 Financial Analysis

Financial features such as credit limit, bill amounts, payment amounts, and payment delays were analyzed to understand their relationship with default behavior.

- `LIMIT_BAL`: KDE plot of credit limit against the target variable **`next_month_default`** shows that customers with lower credit limits (below 1,00,000) have a higher percentage of defaults compared to those with higher credit limits. Also data is highly skewed towards lower credit limits, inducting outliers with high values but since these can be so meaningful in financial context representing (premium customers), we decided to keep them for further analysis.
- `BILL_AMT1 to BILL_AMT6`: While analyzing the monthly bill amounts, we observed that for each month, defaulters tend to have lower bill amounts compared to non-defaulters. This suggests that defaulters may be using their credit cards less or are unable to manage higher spending.
- `AVG_Bill_amt`: The average bill amount across the six months shows a similar trend, with defaulters having lower average bills compared to non-defaulters and this feature also highly skewed towards left side leading to more than 2000 outliers, but again we decided to keep them as they represent meaningful financial behavior.
- `PAY_AMT1 to PAY_AMT6`: The repayment amounts for defaulters are generally lower than those for non-defaulters. This indicates that defaulters may be making smaller payments or struggling to meet their repayment obligations.
- `PAY_0 to PAY_6`: The payment delay status shows that defaulters tend to have more instances of delayed payments compared to non-defaulters. This suggests that a history of delayed payments is a strong indicator of potential default.
- `PAY_TO_BILL_ratio`: The ratio of payment to bill amount shows that defaulters tend to have lower ratios compared to non-defaulters. This indicates that defaulters may be struggling to keep up with their bills relative to their payments.

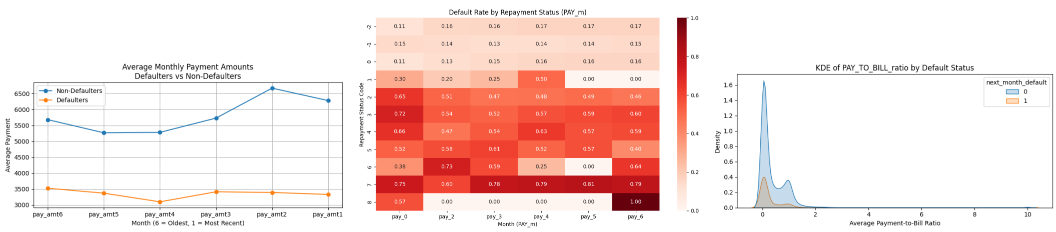

Figure 2: Distributions of Financial Features against Default Status

4 Feature Engineering From Behavioral Analysis

While traditional EDA focuses on understanding the structure and distribution of data, financial and behavioral analysis dives deeper into customer repayment patterns, credit usage behavior, and risk indicators that more directly relate to default likelihood. This type of analysis is crucial in credit risk modeling, as default events are often driven by subtle trends in a customer's financial decisions over time. In this section, we evaluated key behavioral signals such as credit utilization,

bill growth, delay streaks, payment consistency, and repayment-to-bill ratios. These insights were instrumental in designing features that aligned with how lending institutions assess credit risk and customer reliability.

4.1 Credit Utilization Ratios

Credit utilization reflects how much of a customer's available credit limit is being used, which is a key indicator in credit risk scoring. We calculated the following utilization-based features:

- **Util_0:** Ratio of the most recent month's bill (`Bill_amt1`) to the credit limit. Indicates immediate credit pressure.
- **UTIL_max:** Highest utilization across six months. Captures peak credit usage and potential financial stress points.
- **UTIL_avg:** Average utilization across six months. Reflects long-term credit behavior.

Higher utilization values, especially peak and average, were commonly observed in defaulters, indicating risk due to over-leveraging credit.

4.2 Repayment and Billing Volatility

To capture consistency in customer financial behavior, we calculated standard deviations for repayment and billing amounts across six months:

- `pay_std`: Measures how much monthly repayments vary. High values indicate irregular payments and financial instability; low values suggest disciplined repayment habits.
- `bill_std`: Measures variation in monthly bills. High values reflect erratic spending behavior, while low values point to steady and controlled usage.

Defaulters generally exhibited higher volatility in both measures, making these features valuable for risk prediction.

4.3 Delay-Based Features

To capture customer payment discipline, we engineered features that quantify both the frequency and pattern of delayed payments:

- **overdue_cnt:** Total number of months with delayed payment (where delay ¿ 0). More delays often signal higher default risk.

- **on_time_ratio:** Proportion of months paid on time or in advance. A high ratio indicates good repayment behavior.
- **num_months_late:** Counts the number of months with significant delay (PAY_x ≥ 1). Provides another view of repayment discipline.
- **longest_delay_streak:** Maximum number of consecutive months with payment delay. Longer streaks are strongly associated with future defaults.

4.4 Payment Trend Features

Understanding how payments evolve over time helps identify financial stress or improving discipline:

- **payment_slope:** Linear trend (slope) of monthly payments. A negative slope suggests declining repayment ability, which may indicate rising risk.
- **pay_ewm_sum:** Weighted sum of recent payments, giving more weight to recent months. Captures short-term repayment behavior effectively.

4.5 Billing Pattern Change

- **bill_growth_1:** Measures the change in bill amount from month 2 to month 1. A sharp increase may suggest sudden spending spikes, which could precede financial strain.

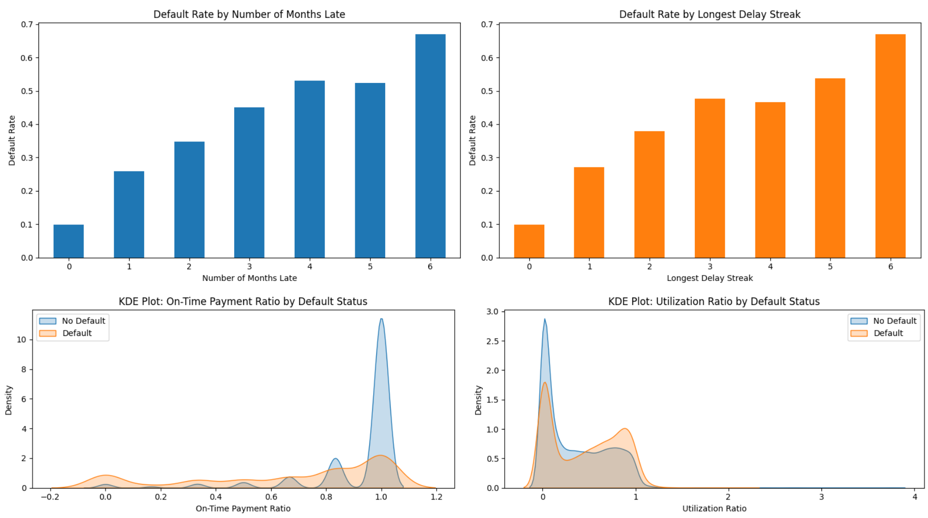

Figure 3: Behavioral Features Engineered Based on Behavioral risk Based Analysis

5 Feature Selection

After engineering a comprehensive set of features, we performed feature selection to identify the most impactful variables for our classification task. This step is crucial to reduce noise, improve model interpretability, and enhance generalization.

5.1 Feature Importance Analysis

We have analyzed feature importance using correlation coefficients, mutual information scores, and tree-based feature importance metrics. After evaluating the impact of each feature on the target variable `next_month_default`, we identified the best representative key features that significantly contribute to predicting credit card defaults.

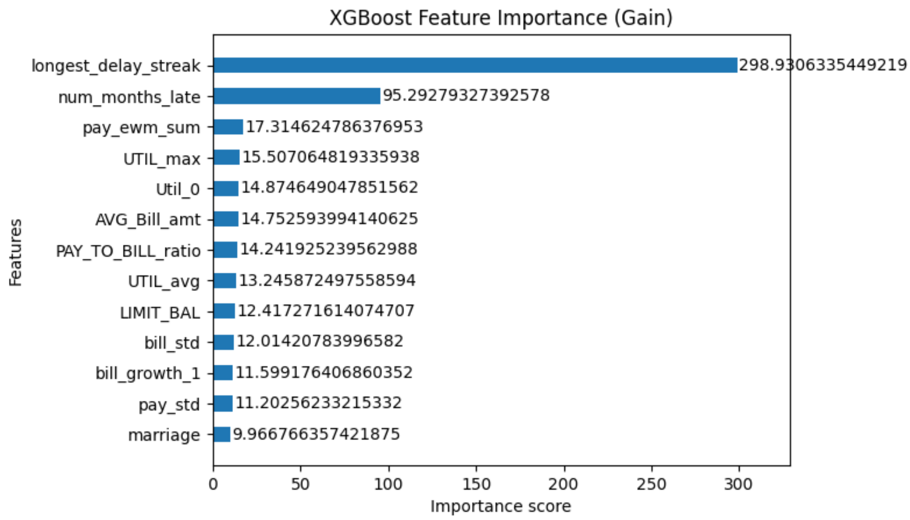

Figure 4: Feature Importance from XGBoost Model

- `marriage` – Marital status has a moderate correlation with default risk.
- `LIMIT_BAL` – Credit limit is a strong predictor of financial behavior.
- `AVG_Bill_amt` – Average bill amount reflects overall credit usage.
- `PAY_TO_BILL_ratio` – Indicates how well customers manage their bills relative to payments.
- `longest_delay_streak` – Captures the pattern of delayed payments, which is critical for risk assessment.
- `num_months_late` – Total months with significant delay directly correlates with default likelihood.
- `Util_0`, `UTIL_avg`, `UTIL_max` – Credit utilization ratios are strong indicators of financial stress.

- `overdue_cnt` – Number of overdue months is a direct measure of repayment discipline.
- `on_time_ratio` – Proportion of on-time payments indicates reliability.
- `bill_growth_1` – Sudden changes in billing patterns can signal potential issues.
- `bill_std, pay_std` – Variability in billing and payment amounts reflects financial stability.
- `pay_ewm_sum` – Recent payment trends provide insights into current financial behavior.

These features were selected based on their strong correlation with the target variable and their ability to capture different aspects of customer behavior that contribute to default risk. when combined, these features provide a comprehensive view of customer financial health, credit usage patterns, and repayment discipline which we have proved through correlation analysis which shows that these features are highly correlated with the target variable `next_month_default`.

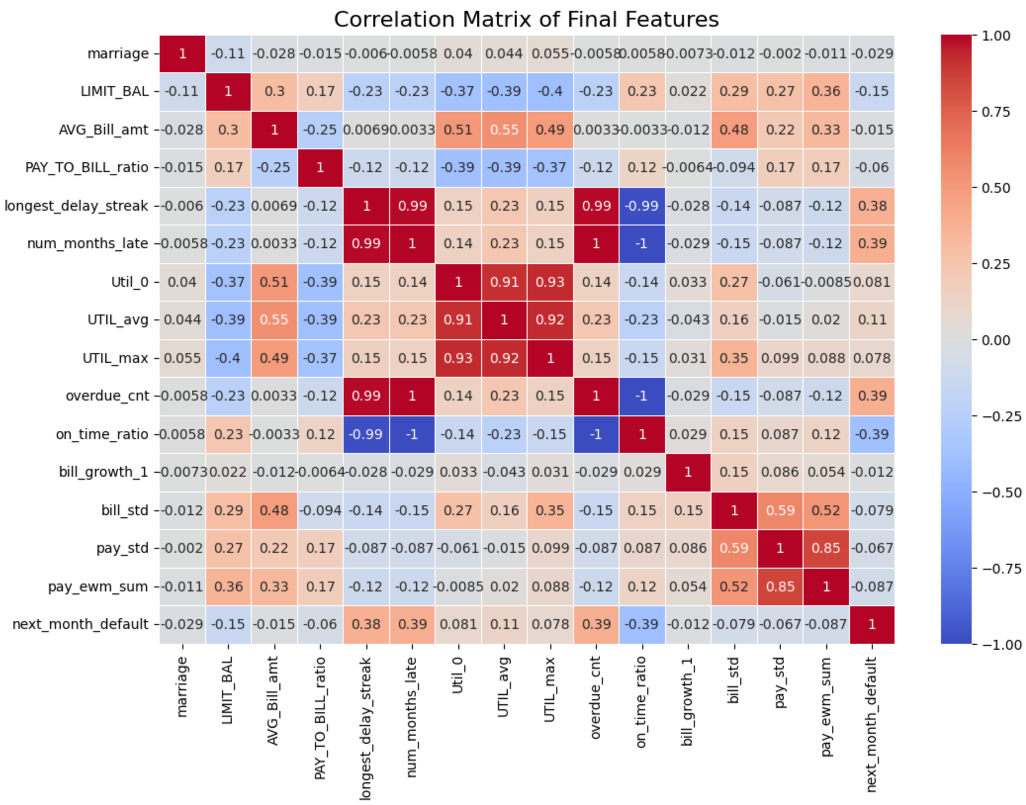

Figure 5: Correlation Heatmap of Selected Features with Target Variable

6 Class Imbalance Handling and Model Training

Multiple machine learning models were trained and evaluated to predict whether a customer will default in the upcoming month. Each model was tested with different strategies for handling class imbalance and optimizing performance using validation data. This section outlines the models used and compares their results.

6.1 Class Imbalance Handling

Class imbalance is a significant challenge in credit default prediction, as the number of defaulters is typically much lower as shown in the script. To address this, we have used two primary techniques:

- **Class Weights:** Most models were trained with `class_weight='balanced'`, which adjusts the loss function to give more importance to the minority class (defaulters). This helps improve recall and F_2-score without oversampling.
- **SMOTE (Synthetic Minority Over-sampling Technique):** SMOTE was applied to oversample the minority class by generating synthetic samples. This technique helps balance the dataset but can lead to overfitting if not used carefully.

6.2 Logistic Regression

Logistic Regression was used as a baseline linear model due to its interpretability and speed. Several variations were tested:

- **Class-Weighted Logistic Regression:** The model was trained from inbuilt standard library `sklearn.linear_model.LogisticRegression` with `class_weight='balanced'` to give more importance to the minority class (defaulters). This approach helped improve recall and F_2-score compared to standard training.
- **SMOTE + Logistic Regression:** The training set was oversampled using SMOTE to synthetically generate additional defaulter samples. Although recall improved, the overall F_2-score dropped slightly due to reduced model calibration, making this method less optimal.

Logistic Regression served as a strong baseline, but its performance plateaued around an F_2-score of 0.5797, leading us to explore non-linear models that could capture more complex relationships.

6.3 Random Forest

Random Forest, a tree-based ensemble model, was tested to capture non-linear patterns and feature interactions that logistic regression could not. Several variations were explored:

- **Class-Weighted Random Forest:** The model was trained with `class_weight='balanced'` to adjust for class imbalance. This improved F_2-score significantly over logistic regression and provided better recall with strong AUC performance.

- **SMOTE + Random Forest:** Oversampling with SMOTE followed by Random Forest training was tested, but the model's recall dropped sharply and the F_2-score decreased. This indicated overfitting to synthetic examples, and the approach was discarded.

- **Feature-Optimized Random Forest:** A version using only top-ranked features (based on importance) showed slightly better generalization and a more interpretable model without hurting performance.

Random Forest achieved a strong F_2-score of 0.5925 and recall of 0.7931 on the validation set, outperforming Logistic Regression and demonstrating its ability to handle complex interactions in the data.

6.4 LightGBM

LightGBM, a gradient boosting framework optimized for speed and efficiency, was trained as a next step. It is particularly well-suited for tabular data and handles categorical-like features and missing values effectively.

- **Baseline LightGBM:** Trained with `class_weight='balanced'` and default hyperparameters. It has shown oversampling towards the minority class due to its inherent handling of class imbalance.

- **SMOTE + LightGBM:** SMOTE oversampling was applied before training. This approach improved F_2-score and recall, leading to a better balance between precision and recall compared to Random Forest.

- **Feature-Pruned LightGBM:** After identifying low-importance features, we removed them and retrained the model. This improved both F_2-score and recall slightly, while keeping the model simpler and more focused.

LightGBM achieved an F_2-score of 0.5753 and recall of 0.7526, demonstrating strong performance on the validation set. Its speed and efficiency made it a strong candidate for further optimization.

6.5 XGBoost

XGBoost, another powerful gradient boosting framework, was trained to capture complex relationships in the data. It is known for its speed and performance on structured datasets. But since it is highly prone to overfitting, it was trained with careful hyperparameter tuning and feature selection. Although it is speedy, it failed to outperform Random Forest in terms of F_2-score.

Now, After training all of these models, we compared their performance on the validation set using key classification metrics: Accuracy, Recall, F1 Score, F_2-Score, and AUC-ROC. The goal was to first decide which metric to prioritize based on business objectives, and then select the best model accordingly.

7 Model Comparison and Selection

After evaluating multiple models on validation data, we compared their performance using key classification metrics: Accuracy, Recall, F1 Score, F_2-Score, and AUC-ROC. The goal was to first decide which metric to prioritize based on business objectives, and then select the best model accordingly.

Model	**Precision**	**Recall**	**F1 Score**	**AUC-ROC**	**F_2-Score**
Logistic Regression	0.5954	0.7734	0.4214	0.7496	0.5797
Random Forest	0.5925	0.7931	0.4295	0.7529	0.5925
LightGBM	0.6123	0.7526	0.4251	0.7360	0.5753
XGBoost	0.4988	0.8565	0.3944	0.7378	0.5832

Table 1: Model Comparison on split Validation Data

7.1 Threshold Optimization

To align the model's predictions with our primary evaluation metric (F_2-score), we did not use the default probability cutoff of 0.5. Instead, we plotted the precision–recall curve on the validation set and computed F_2-score at each candidate threshold. The F_2-score is defined as

$$F_2 = \frac{(1 + 2^2)\,\text{Precision} \times \text{Recall}}{2^2\,\text{Precision} + \text{Recall}},$$

which weights recall twice as heavily as precision. To better align with business priorities, we optimized the classification threshold by maximizing the F_2-score on the validation set. The F_2-score places more weight on recall than precision, ensuring that more actual defaulters are correctly identified, even if it means allowing a few more false positives.

A grid of thresholds from 0.1 to 0.9 (in steps of 0.01) was evaluated. For each threshold, predictions were generated from model probabilities, and the corresponding F_2-score was calculated. The threshold that yielded the highest F_2-score was then selected for final predictions. This approach allowed the model to better reflect the bank's risk appetite by emphasizing the identification of high-risk customers.

While the default threshold of 0.5 is suitable for balanced datasets, it does not account for the class imbalance in credit default prediction. By optimizing the threshold based on F_2-score, we ensured that the model was tuned to minimize missed defaulters while controlling false alarms. We found `threshold = 0.20` to be the optimal threshold for Random Forest model, which achieved the highest F_2-score of 0.5925 on the validation set.

7.2 Risk Based Model Selection

In our risk-based evaluation framework, we prioritized recall (minimizing missed defaulters) and F_2-score above all, leading to the selection of Random Forest as the final model for three reaons:

1. **Highest F_2-Score:** At the optimized threshold of 0.20, Random forest achieved an F_2 of 0.5925, surpassing all other
2. **Strong Recall and AUC:** With recall = 0.7931 and AUC-ROC = 0.7529, Random Forest demonstrated a strong ability to identify defaulters while maintaining a reasonable false positive rate.
3. **Focused Feature Use:** After pruning low-importance features, the model retained only the most impactful variables, improving interpretability and reducing noise. This aligns with risk-based modeling principles that emphasize key risk indicators.

These factors confirm random Forest as the best choice for production-style predictions on the validation dataset, ensuring maximal defaulter coverage with controlled false alarms.

8 Final Predictions and Conclusion

From the final Random Forest model, we generated probability of default for each customer in unseen `validate_dataset_final.csv.` and this is how probability distribution looks like:

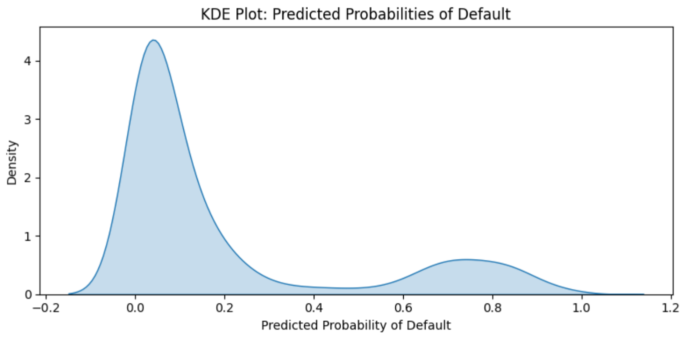

Figure 6: Probability Distribution of Default Probabilies

As we can see, the distribution contains a peak around 0.1 and also a peak around 0.8, indicating that our model is able to identify customers with high risk of default as well as those with low risk. Also our optimal threshold of 0.20 includes customers in middle range, which are neither high risk nor low risk, but can be considered as potential defaulters.

Default (1)	**No Default (0)**
1287	**3729**

Table 2: Final Predictions on Validation Set

The final model's predictions were saved in file `submission_23114104.csv` in the required format [`Customer_ID,next_month_default ( 0 or 1 )`]. In github repository provided below you can find the datasets, model training scripts, and final submission file.

GitHub

Thank You!